Lauryn N. Howlett

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ECOSYSTEM SCIENCE & SUSTAINABILITY PROFILE

Dedicated to understanding and finding solutions to sustainability issues in the ecological sector. I am passionate about helping communities connect to nature and improving conservation efforts. My core competencies include:

- Understanding of current sustainability and climate change-related issues
- Communication and coordination between multiple stakeholders
- Environmental research, data collection, and analysis
- Data analysis and visualization using R and Excel

EDUCATION

Colorado State University (CSU), *Fort Collins, CO* — *December 2024*
Professional Science Master's, **Ecosystem Science and Sustainability, *Sustainable Food Systems*** — **4.0 GPA**

Colorado State University (CSU), *Fort Collins, CO* — *December 2024*
Graduate Certificate, **Water Resources** — **4.0 GPA**

Colorado State University (CSU), *Fort Collins, CO* — *May 2021*
Bachelor of Science, **Ecosystem Science and Sustainability** — **3.84 GPA**
Minor in **Sustainable Energy**
Honors program Cum Laude

CONSERVATION, SUSTAINABILITY AND PROFESSIONAL EXPERIENCE

Research Assistant, *Agricultural Water Quality Program, Colorado State University, Fort Collins, CO* — *May 2024 - October 2025*

- Analyzed water quality data using Excel and R Studio
- Supported the installation and maintenance of automated water samplers for edge-of-field research projects
- Enhanced ability in quality control, storage, and manipulation of large datasets to support research questions
- Integrated the use of technologies, including 3D printing and drone operation, to optimize data collection
- Contributed to public outreach events by presenting the use of technology in research and water quality knowledge

Graduate Teaching Assistant, *Colorado State University, Fort Collins, CO* — January 2024 – May 2024

- Supported instruction and course management for ESS 312: Sustainability Science, collaborating with faculty to enhance student learning experiences
- Evaluated assignments, maintained accurate records in the grade book, and tracked student attendance to ensure smooth course operations
- Provided timely and effective responses to student inquiries, offering guidance on course material, and fostering an engaging academic environment

Land Stewardship Intern, *Colorado Open Lands, Lakewood, CO* — June 2021 - December 2021

- Coordinated and scheduled meetings with landowners, fostering strong relationships and clear communication regarding conservation easements
- Monitored 264 properties, totaling 117,232 acres. across the state of Colorado using both field assessments and satellite imagery to ensure conservation easement compliance
- Created and updated property maps with ArcMap, supporting accurate documentation and reporting
- Developed proficiency with Lens, an online satellite mapping platform, and conducted vegetation health analyses to assess land conditions
- Interpreted conservation easement language to evaluate property restrictions and stewardship requirements

Lab Assistant, *Natural Resource Ecology Laboratory, EcoCore, Fort Collins, CO* — August 2019 – May 2021

- Received and prepared NEON water samples for analysis, following proper protocols and ensuring accuracy
- Assisted in DOC, DIC, and TOC analysis on water samples
- Refurbished bottles and washed other laboratory dishes, maintaining a clean and organized workspace
- Assisted with other laboratory practices and machine maintenance

Research Fellow, *USDA NIFA REEU, CSU, Fort Collins, CO* June 2019 – July 2019

- Performed soil moisture data analysis from an existing tillage dataset in R
- Field research experience assisting a PhD student collect biomass for grassland study
- Analyzed soil health in relation to water holding capacity and different tillage practices
- Participated in professional development workshops
- Presented the results to CSU faculty

COLLABORATIVE RESEARCH AND LEADERSHIP EXPERIENCE

Honors Senior Thesis, *Colorado State University* August 2020 – May 2021

- Mentored by Dr. Jill Baron and committee member Dr. Daniela Cusack
- Analyzed long term silica trends in the Loch Vale watershed in Rocky Mountain National Park (RMNP), Colorado
- Assisted in water sample collection at Loch Vale and Sky Pond
- Conducted a literature review on the importance of silica and similar trends
- Presented my thesis presentation in front of advisors and other faculty members

Senior Capstone Project, *Colorado State University* January 2021 - May 2021

- "Design and Collaborative Implementation of a Pollinator Habitat on a Colorado Front Range Farm"
- Coordinated with Wildlands Restoration to design and execute a volunteer project where we planted seeds and bushes to create a pollinator habitat within a crop field
- Presented research at the Youth Environmental Alliance in Higher Education (YEAH) Conference at Colorado State University

TECHNICAL AND ANALYTICAL SKILLS

- Develop succinct, professional reports and present results effectively in various settings
- Consistently deliver high-quality results by organizing tasks and meticulously managing details, even when working under tight deadlines
- Mastery in Microsoft Office 365, including Excel and Word processing systems
- Experience using R Studio for data analysis and visualization
- Demonstrated strength in written report creation and oral communication
- Capability to conduct fieldwork and follow proper protocol
- Ability to navigate in remote areas using GPS software and a compass

RELEVANT COURSEWORK

GIS and Data Analysis in Water Resources, *CSU, Fort Collins, CO* Spring 2024

- Improved GIS skills for natural resource management
- Improved skills using ESRI products

Water Law for Non-Lawyers, *CSU, Fort Collins, CO* Fall 2024

- Learned the fundamentals of international and national water governance and management
- Expanded knowledge of water governance and law in the Western United States, including Colorado

Environmental Data Science Applications, *CSU, Fort Collins, CO* Fall 2023

- Introduction to R programming
- Developed skills to import and manipulate open-source datasets
- Enhanced abilities in data analysis and linear modeling
- Created a personal website using quarto

Earth Systems Ecology Spring 2021

- Connected ecosystem principles to global systems and how they interact
- Completed writing an NSF grant proposal